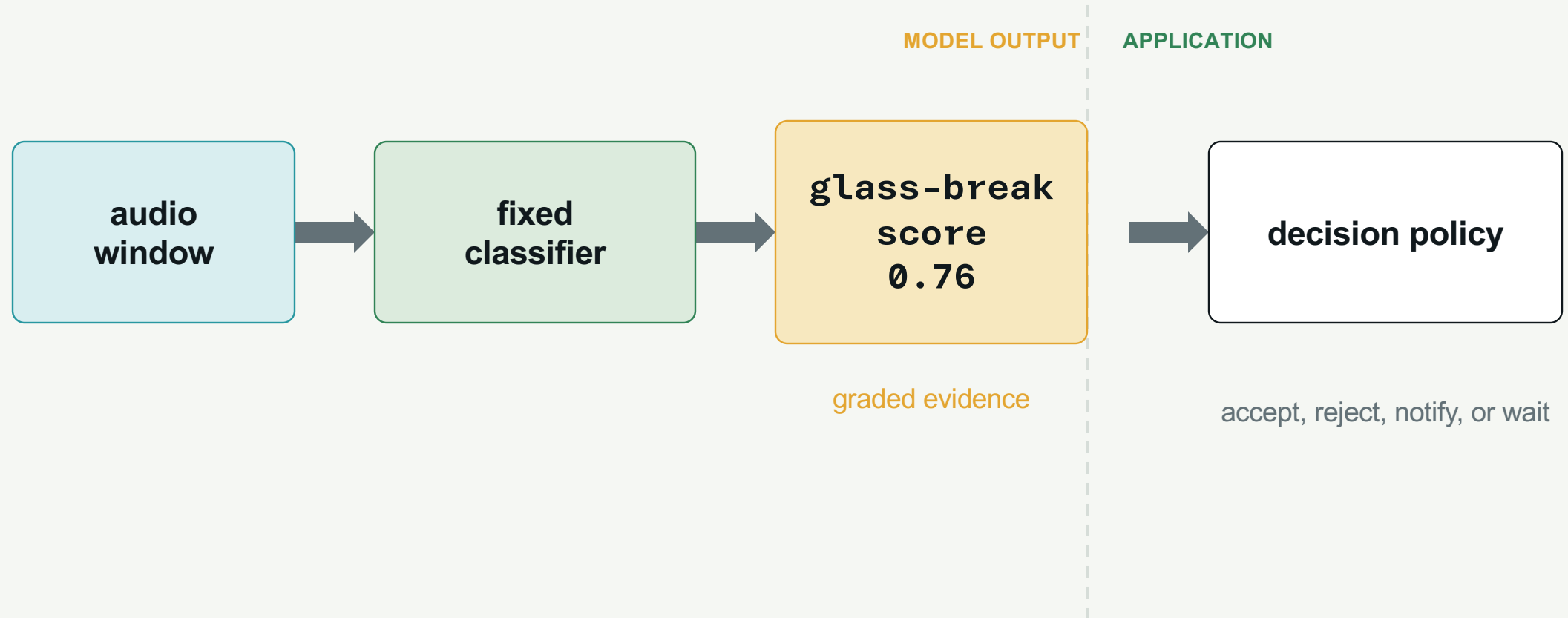


Classification Thresholds and Operating Points

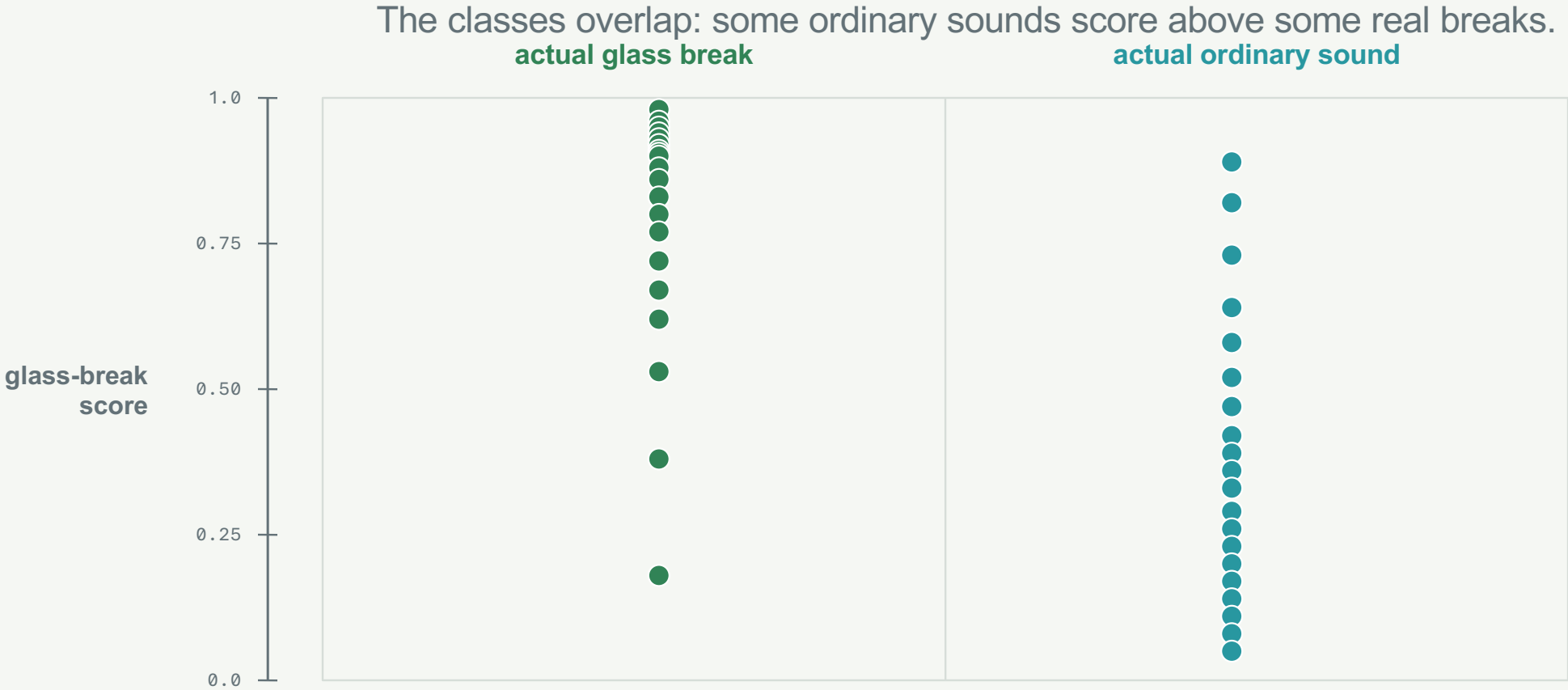
Turning model scores into decisions



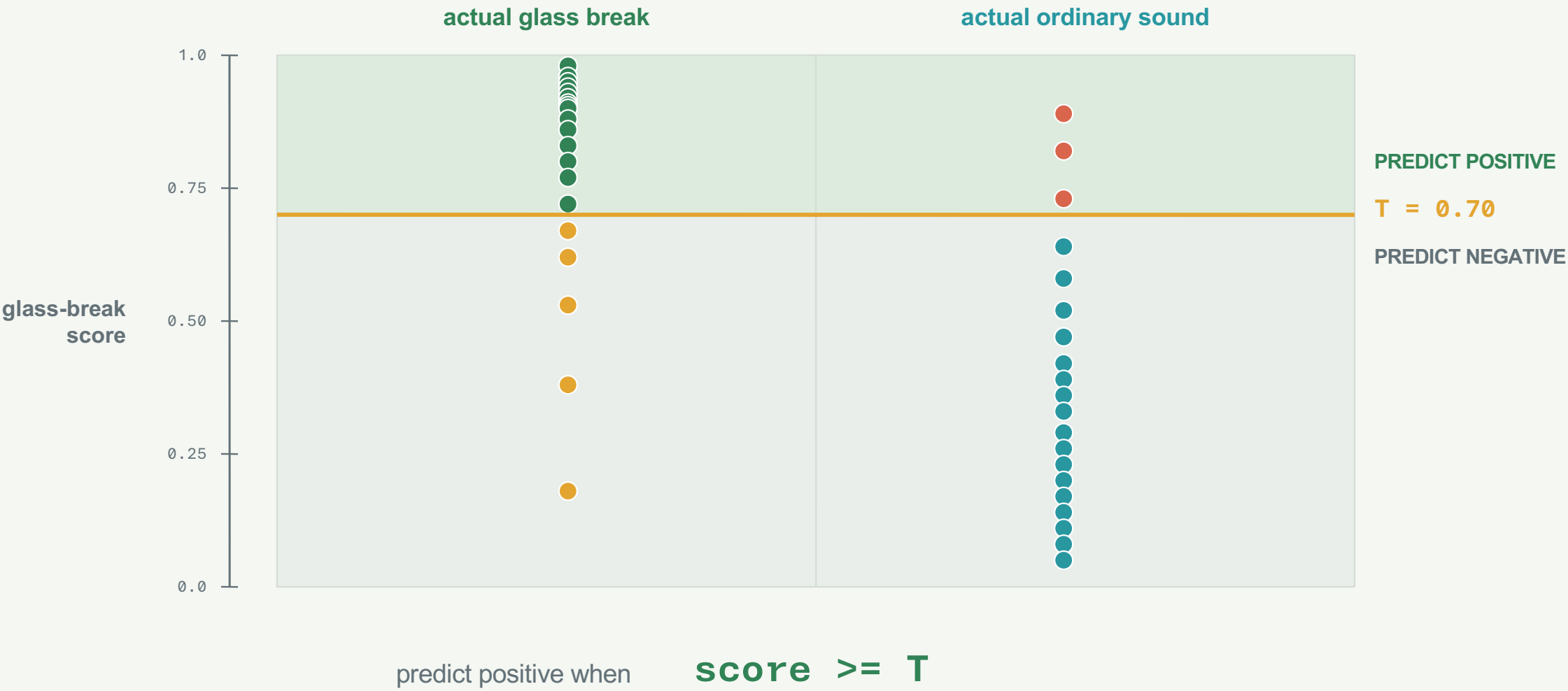
Scores and Decisions



Score Distributions



Decision Threshold



Application Consequences

PHONE NOTIFICATION

A person reviews the alert
before acting.

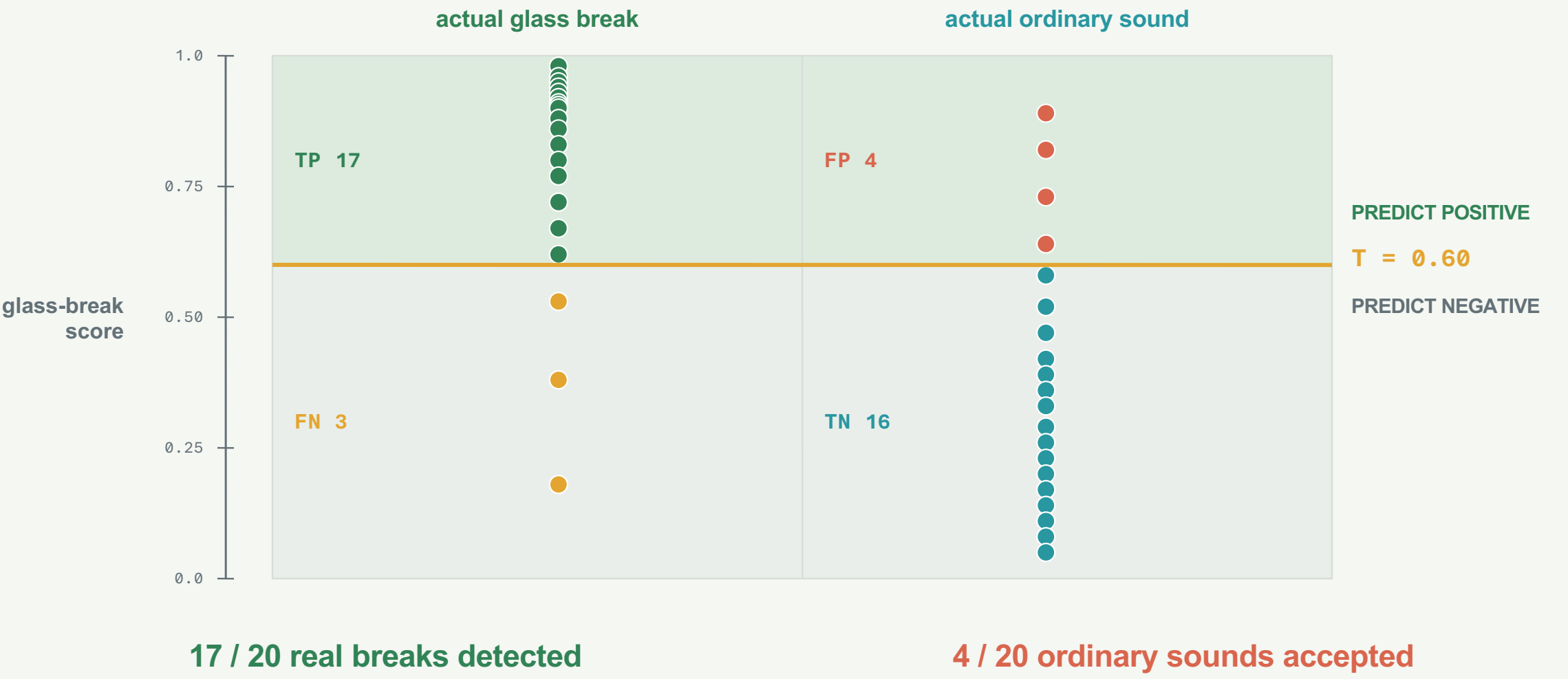
AUTOMATIC RESPONSE

The system acts immediately
without human review.

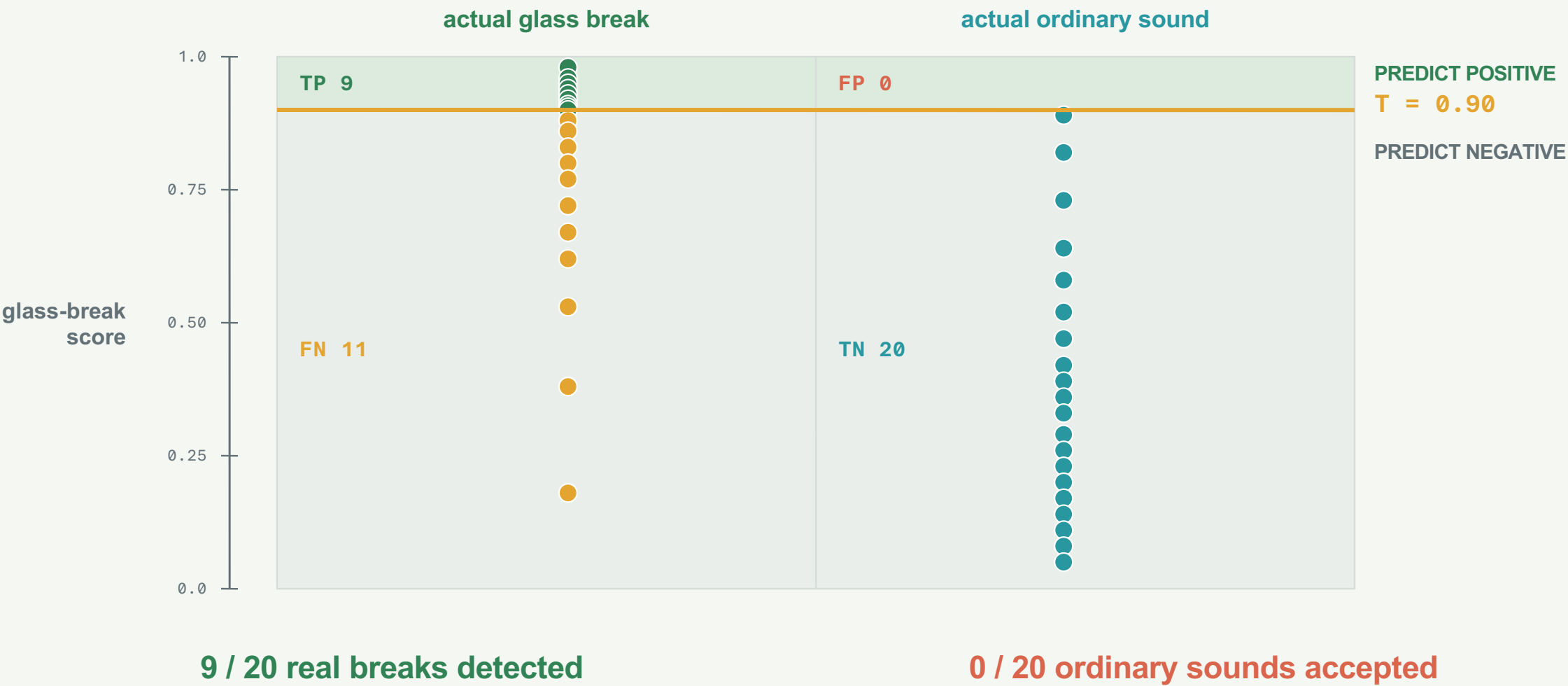
Which error is more costly for each application?

Talk with your neighbor before choosing a threshold.

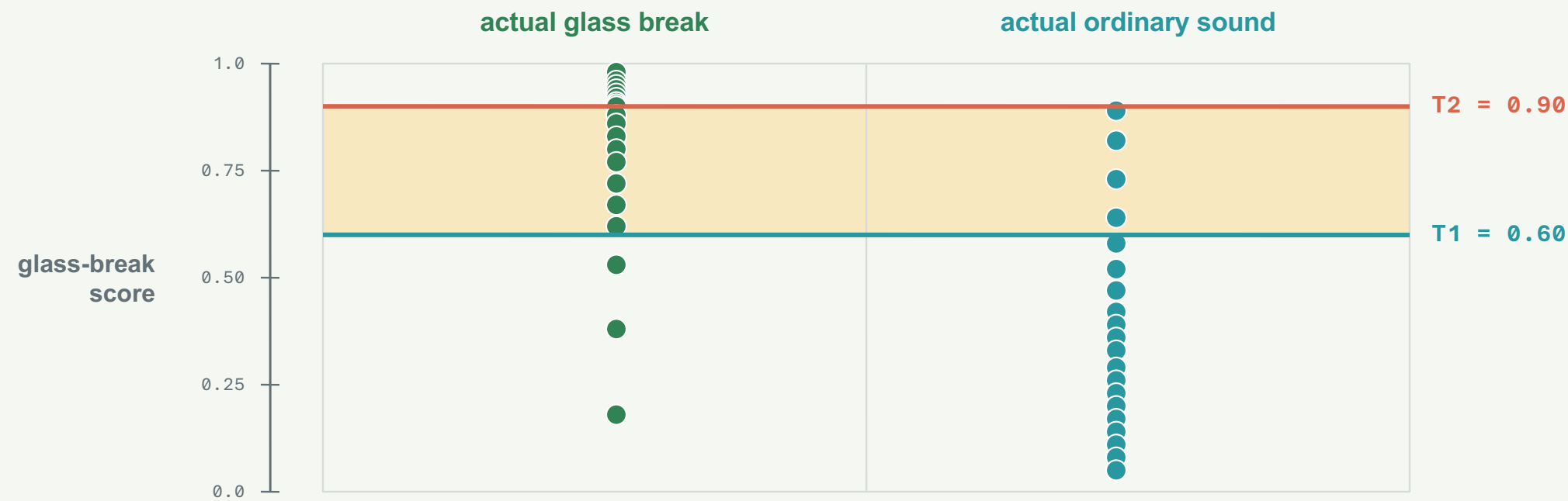
Lower Decision Threshold



Higher Decision Threshold



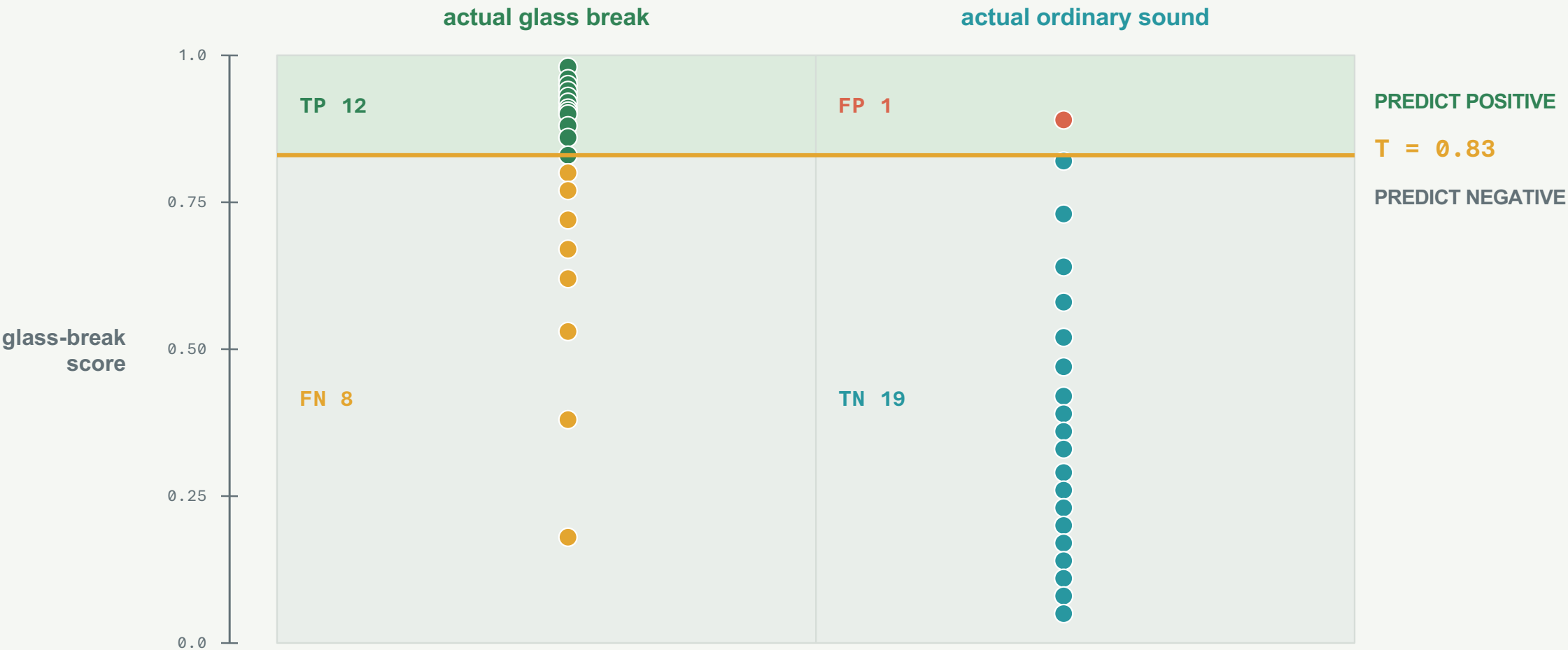
Threshold Monotonicity



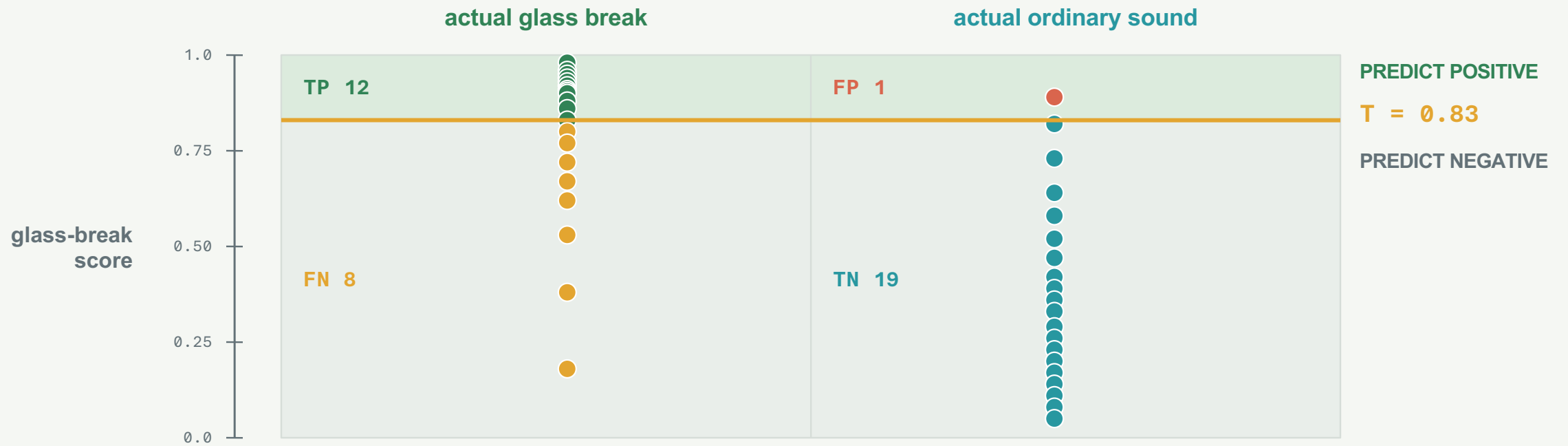
Raising the threshold moves the boundary upward; the accepted set can only shrink.

as T increases	predicted positives	TP and recall	FP and FPR	precision has no monotonic guarantee
	21 -> 9	17 -> 9	4 -> 0	

Binary Decision Outcomes



Precision and Recall at One Threshold



among accepted predictions

$$\text{precision} = 12 / (12 + 1) = 0.92$$

How often is an accepted alert right?

among actual positives

$$\text{recall} = 12 / (12 + 8) = 0.60$$

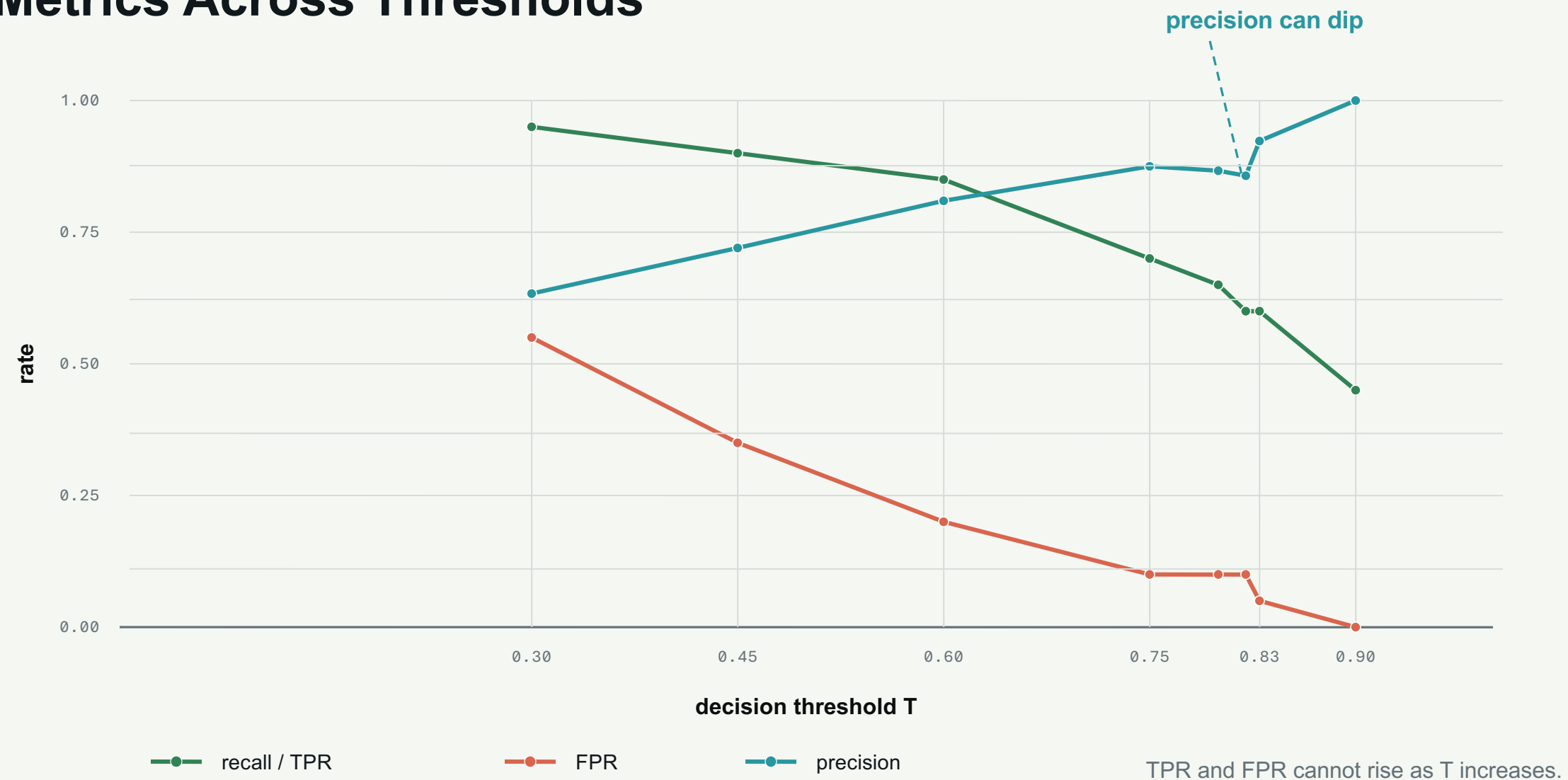
How often is a real break detected?

Threshold Sweep

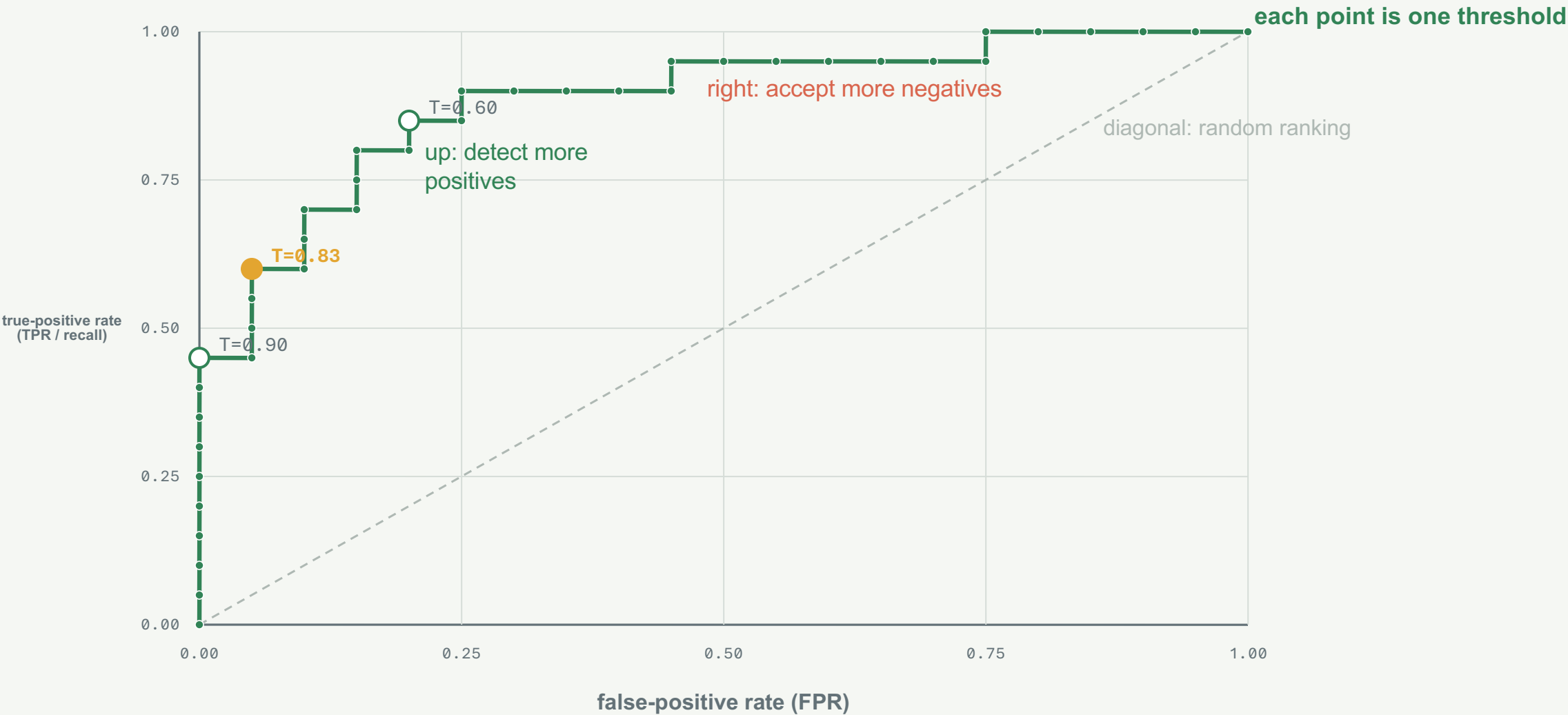
threshold	TP	FP	recall / TPR	FPR	precision
0.30	19	11	0.95	0.55	0.63
0.45	18	7	0.90	0.35	0.72
0.60	17	4	0.85	0.20	0.81
0.75	14	2	0.70	0.10	0.88
0.80	13	2	0.65	0.10	0.87
0.82	12	2	0.60	0.10	0.86
0.83	12	1	0.60	0.05	0.92
0.90	9	0	0.45	0.00	1.00

same fixed model + same 40 labeled clips + different threshold

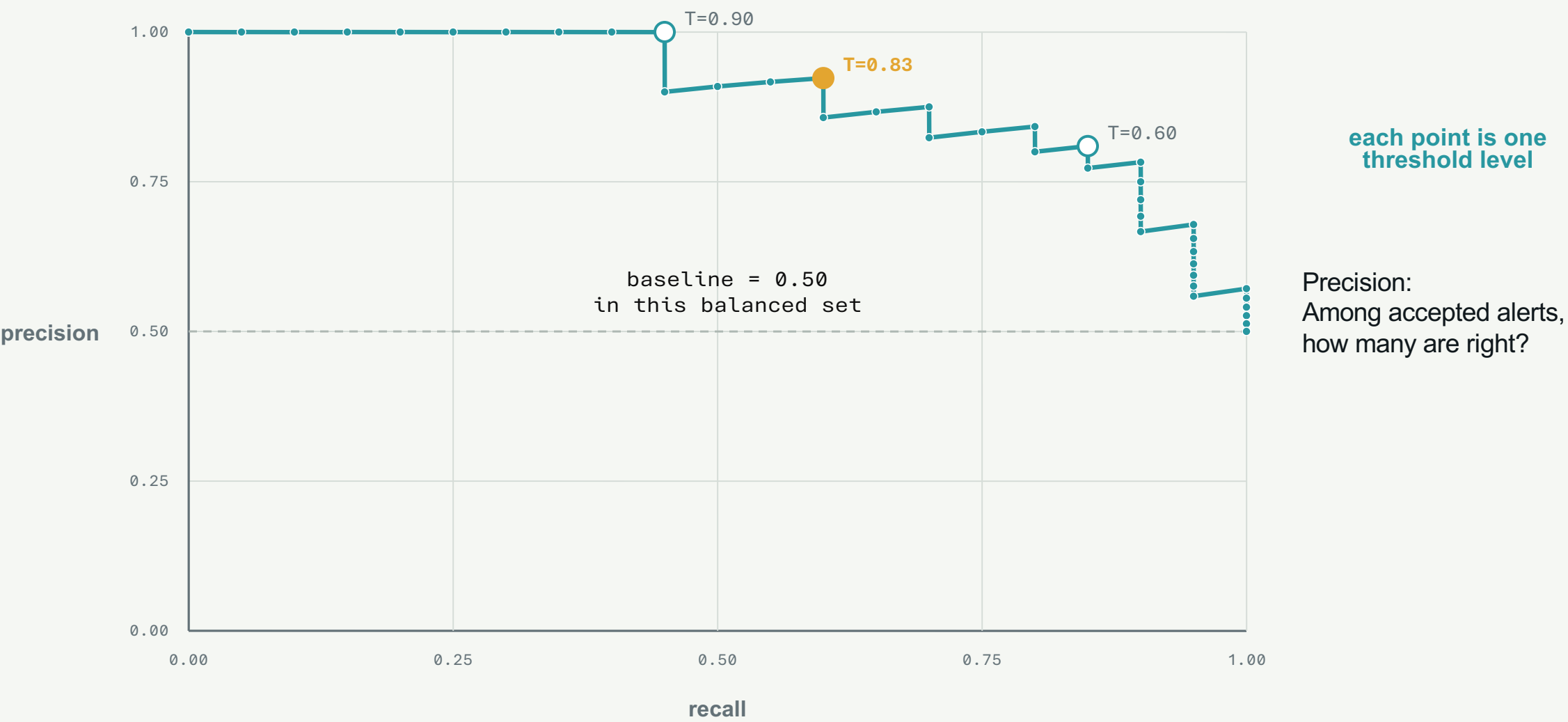
Metrics Across Thresholds



Receiver Operating Characteristic

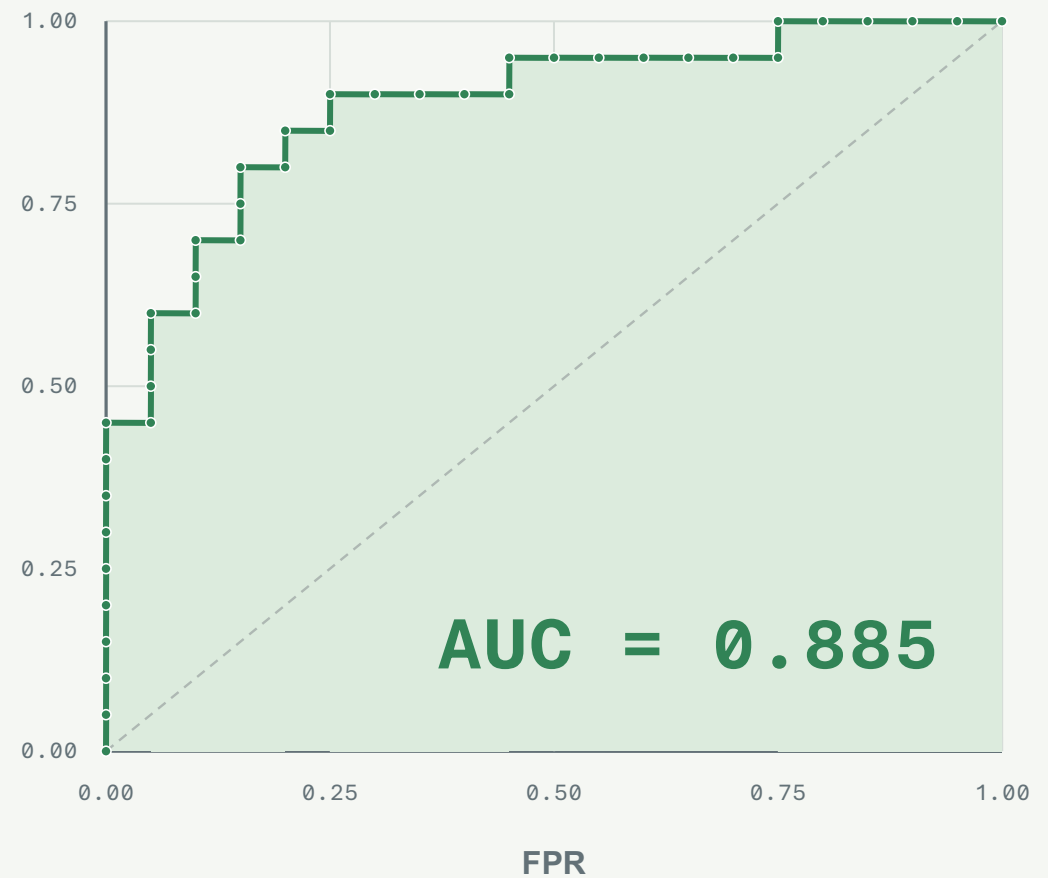
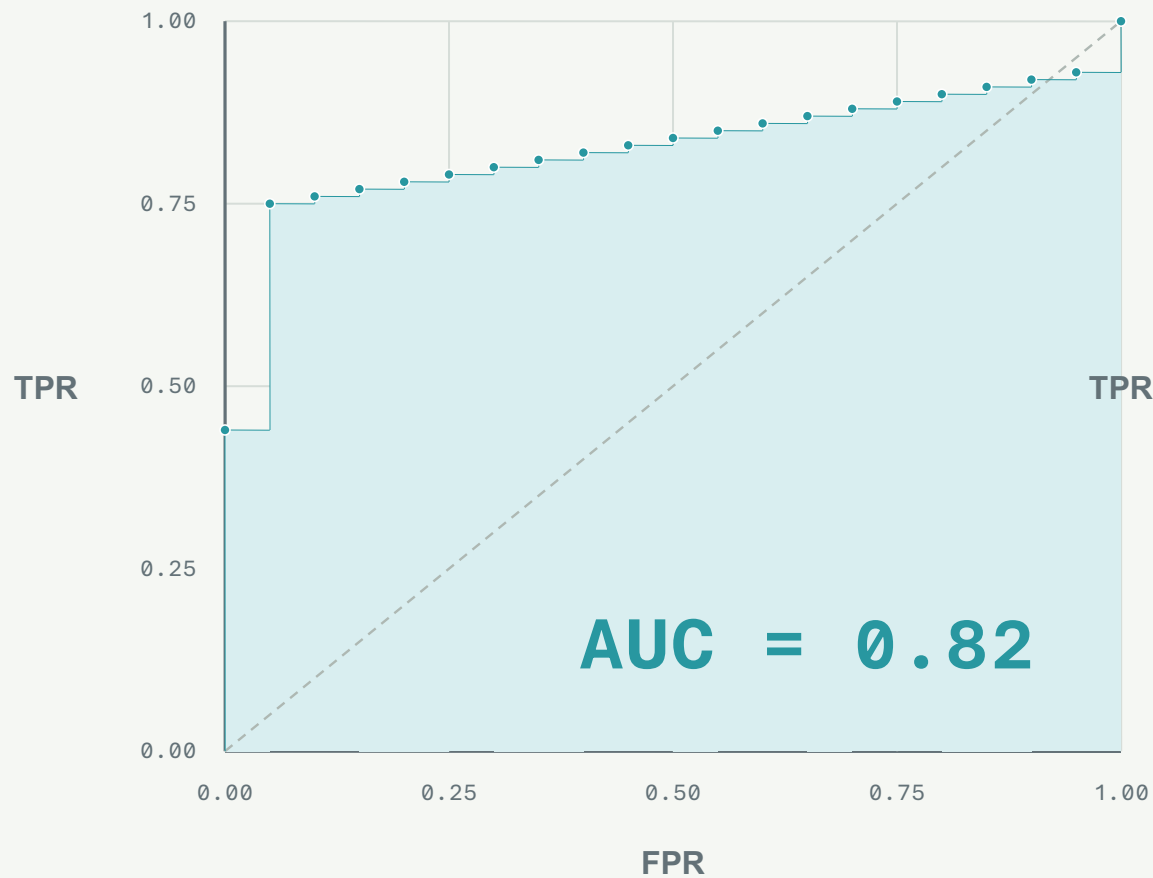


Precision-Recall Curve



Area Under the ROC Curve

$$AUC = \frac{354 \text{ correctly ordered positive-negative pairs}}{400 \text{ total pairs}} = 0.885$$



Operating Requirements

Application goal: automatic response with limited false activations

example limit

**No more than 1 false
activation**

in 20 ordinary validation clips



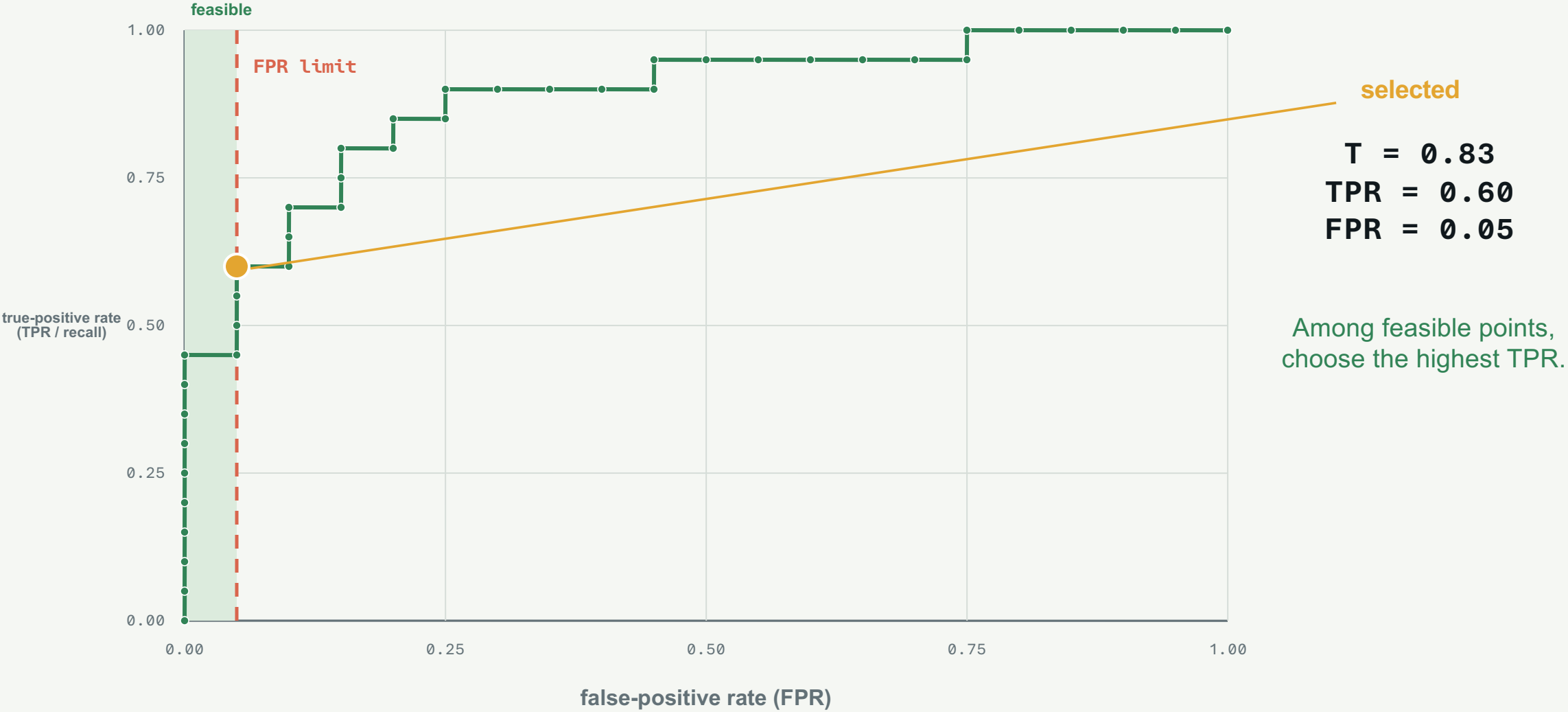
measurable constraint

$FPR(T) \leq 0.05$

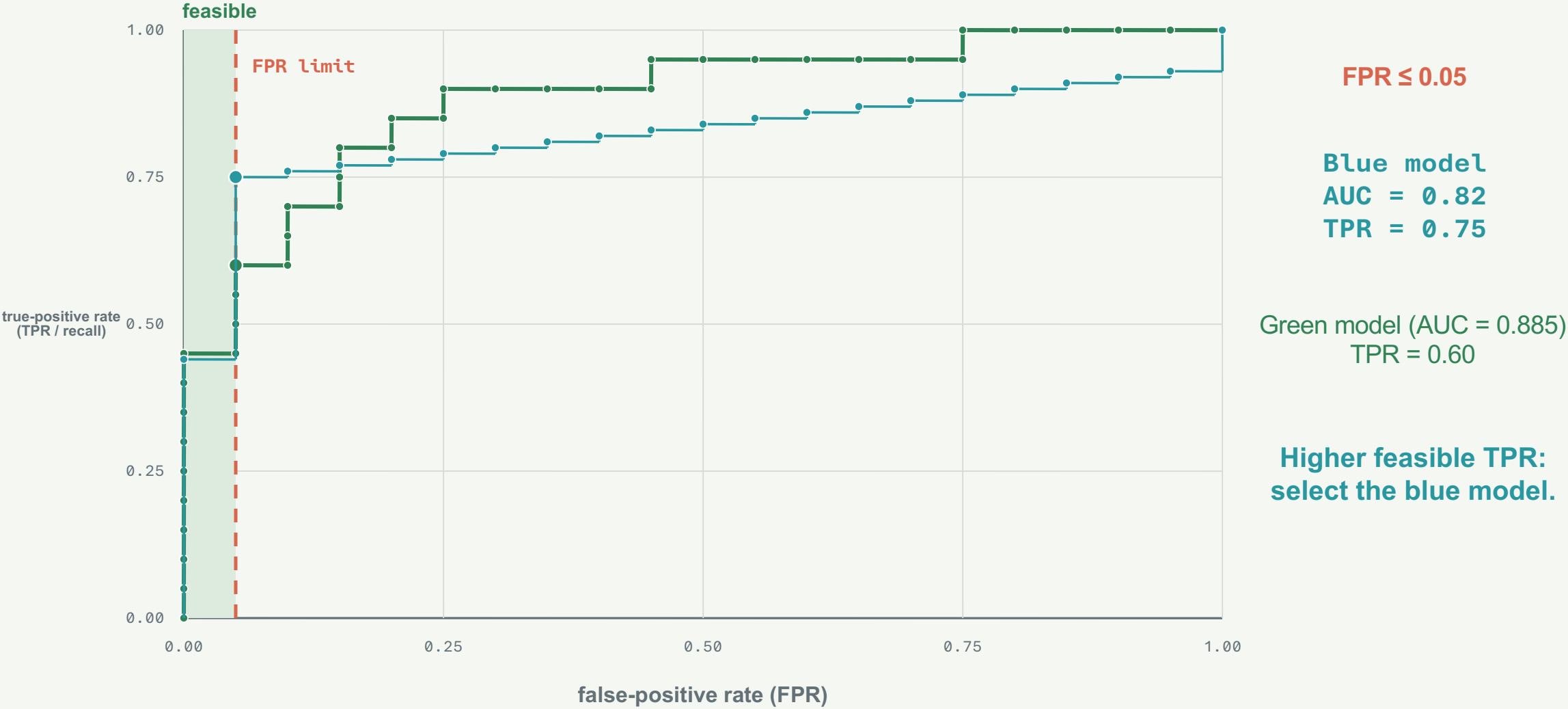
candidate thresholds must satisfy
this limit

Choose T^* to maximize $TPR(T)$ subject to $FPR(T) \leq 0.05$

Constrained Operating Point Selection

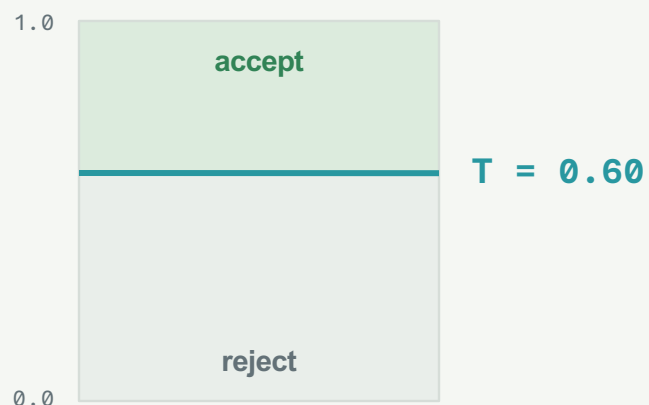


Selecting Between Models at One Requirement



Application-Specific Operating Points

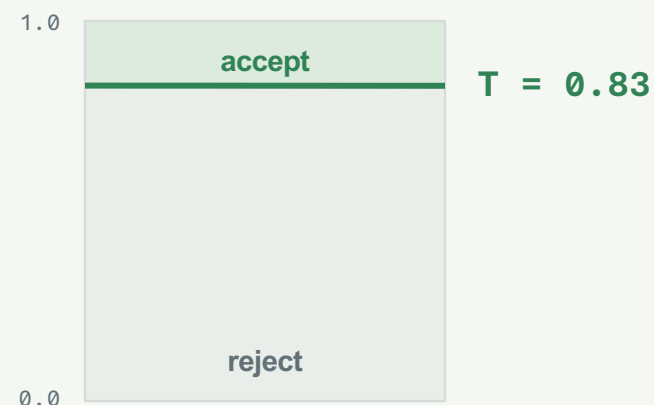
PHONE NOTIFICATION



TPR 0.85 | FPR 0.20

More alerts; a person reviews them.

AUTOMATIC RESPONSE



TPR 0.60 | FPR 0.05

Fewer false alarms; more real events missed.

The model is identical. The operating policy is different.

Prevalence and Precision

same TPR = 0.60 and FPR = 0.05

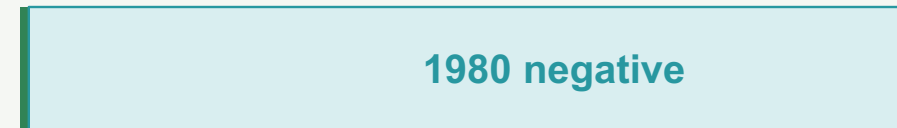
balanced evaluation set



12 TP + 1 FP

precision = 12 / 13 = 0.92

thought experiment: 1% positive windows



20 positive

12 TP + 99 FP

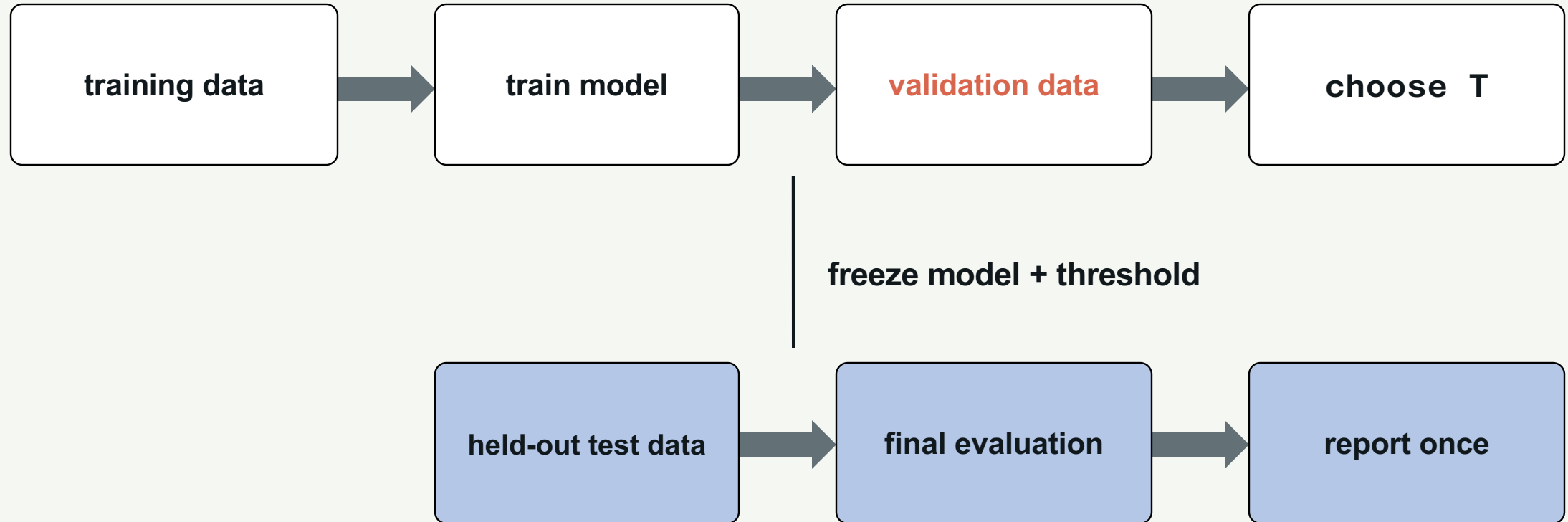
precision = 12 / 111 = 0.11

Window prevalence changed. The class-conditional rates did not.

Long background recordings are needed to estimate false alarms per hour.

Threshold Selection Uses Validation Data

Note: Choose T on the training. Do not choose T on the final test set.



Deployed Decision Policy

Changing any row creates a different evaluated policy.

model version	glass-audio-v3
positive class	glass break
score	glass-break output
accept rule	score ≥ 0.83
validation evidence	TPR 0.60 FPR 0.05
fallback	no automatic action